

# BIPIN ADHIKARI

Machine Learning Engineer

✉ [bipin.adhik@gmail.com](mailto:bipin.adhik@gmail.com)  [in/bipinadhikari101](https://github.com/bipinadhikari101)  [adhikari.pages.dev](https://adhikari.pages.dev)



## Professional Summary

Machine Learning Engineer with Masters in Data Science and hands-on experience in applying ML techniques (LLM, CNN, GNN, etc) to real-world systems. Combined with **MLOps experience** (Docker Swarm, Jenkins, GitHub Actions) from Open Air Interface.

## Experience

### CI and ML Developer

Feb 2024 – Sep 2024

*Open Air Interface*

*Biot, France*

- Trained ML models to **predict faulty code segments and potential errors** based on past logs and past git diffs, providing debugging assistance.
- Implemented **performance monitoring** (CPU, RAM, network, disk usages) and data collection of testing pipelines across multiple servers.
- Simplified and combined old deployment methods so different tests could run simultaneously on different servers.
- Implemented the transition of CI testing pipelines from hard-coded Python scripts to Docker Swarm and Jenkins.
- **Tools:** Scikit-Learn, PyTorch, Pandas, Docker, Jenkins, Git

### Research Project Assistant (LLM / Chatbot)

Oct 2023 – Feb 2024

*Eurecom / Prof Raphael Troncy*

*Biot, France*

- Developed a LLM chatbot for the Odeuropa.eu project, handling a **dataset of 400 million** olfactory related sentences.
- Rigorously Tested and compared multiple embedding models to improve chatbot response time 250ms → 50ms.
- Implemented RAG, memory, reasoning, and multistage LLM prompting to enable LLM to think and understand the problem then find a relevant and accurate solution.
- **Tools:** Langchain, Huggingface, OpenAI GPT 3.5, ChromaDB, Pandas, Knowledge Graph, RAG, react.js

## Projects

### Detecting Faulty Machines Through Sound Anomalies | w/ Prof Pietro Michiardi

Feb 2023 – May 2023

- Developed a semi-supervised model to detect faulty machine sounds using only normal sound data for training.
- Applied DBScan for anomaly detection and CNNs for feature extraction, achieving 98% accuracy on test data.
- **Tools:** DBScan, CNN, ftt, Kaggle, NumPy, Librosa, Matlab

### Graph Neural Network Confidence Testing | w/ Prof. Maurizio Filippone

May 2023 – Aug 2023

- Studied confidence calibration of various GNNs, comparing findings across multiple architectures.
- Tested proposed solutions from research papers on Bayesian GNNs, Graph Attention Networks, Graph Isomorphism Networks, and Graph Convolution Networks to evaluate effectiveness.

## Skills

Machine Learning: PyTorch, TensorFlow, Transformers, CNNs, GNN, Langchain

**LLM and NLP:** Fine Tuning, Few-shot, Lora, Qlora, RAG, Chain Of Thought, BLUE, ROUGE evaluation

**MLOps & DevOps:** Docker Swarm, Kubernetes, CI/CD Pipelines, GitLab/Jenkins, AWS, Performance Monitoring

**Programming & Data:** Python, JavaScript/Node.js, SQL, NoSQL, VectorDBs

**Languages:** English (C2), French(B1), Nepali(Native), Hindi(Native)

## Education

### Eurecom | Institut Mines-Télécom

Sep 2022 – Sep 2024

*Master of Science in Data Science*

*Sophia Antipolis, France*

### UCSI University

Jan 2019 – Jan 2022

*Bachelors of Science in Computing*

*Kuala Lumpur, Malaysia*

### Chat with my AI Assistant!

 [adhikari.pages.dev](https://adhikari.pages.dev)

Ask my AI assistant about my projects, experiences, or my passions – it's like ChatGPT, but all about me!